RAFFLES INSTITUTION

2026 YEAR 6 PRELIMINARY EXAMINATION

Higher 2

CANDIDATE NAME

CLASS | 2 | 6 | S | 0 | 2 | B |

PHYSICS **9478/03**

Paper 3 Longer Structured Questions **18 September 2026**

2 hours

You must answer on the question paper.

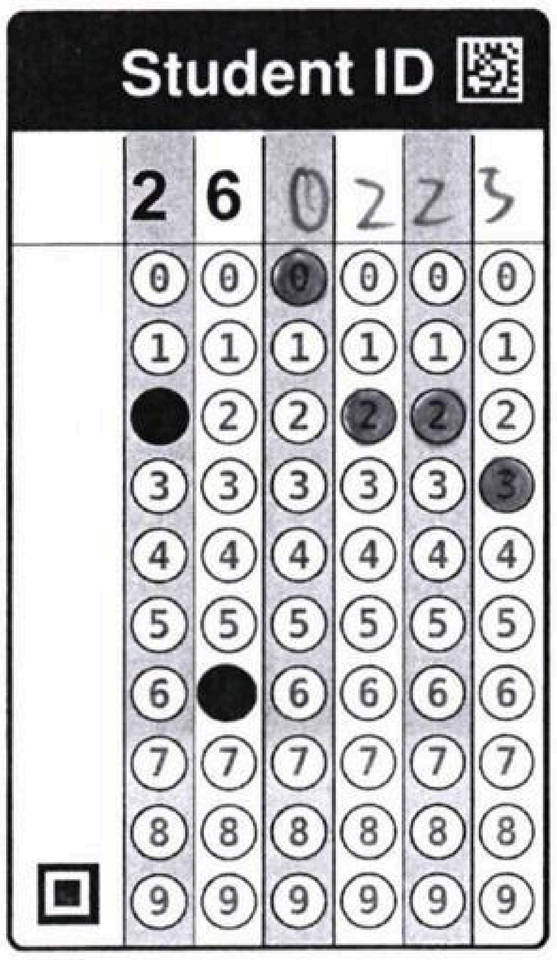
No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question. Circle the question number you attempted and wish to be marked on the cover page.
- Use a black or dark blue pen. You may use a 2B pencil for any diagrams or graphs.
- Write your name and class in the spaces at the top of the page.
- Shade the lozenges in the Student ID box below according to your RI Exam Index No.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- This section carries a total of 75 marks.
- The number of marks for each question or part question is shown in brackets [ ].



For Examiner's Use			
Section A	1	0 /	11
	2	8 /	10
	3	4 /	9
	4	5 /	6
	5	4 /	7
	6	2 /	5
	7	5 /	7
Section B Circle 8 or 9	(8) / 9	8 /	20
Deduction		-1	
Total		35 /	75

This document has **29** pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi\times10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut+\frac{1}{2}at^2$
	v^2	=	u^2+2as
work done on / by a gas	W	=	$p\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T/K	=	$T/°C+273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2\rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$

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displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 \mathrm{I}}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N \mathrm{I}}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n \mathrm{I}$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this Section in the spaces provided.

1 Two spheres A and B of masses 69 g and 250 g respectively are suspended by light inextensible threads from fixed points. When the spheres are at rest, they are in contact with their centres at the same horizontal level, as shown in Fig. 1.1.

The vertical distance from the fixed point of suspension to the centre of sphere A is 1.50 m.

A force F, acting at an angle θ to the vertical, is applied to sphere A, displacing it so that the thread makes an angle of 24° with the vertical, as shown in Fig. 1.2.

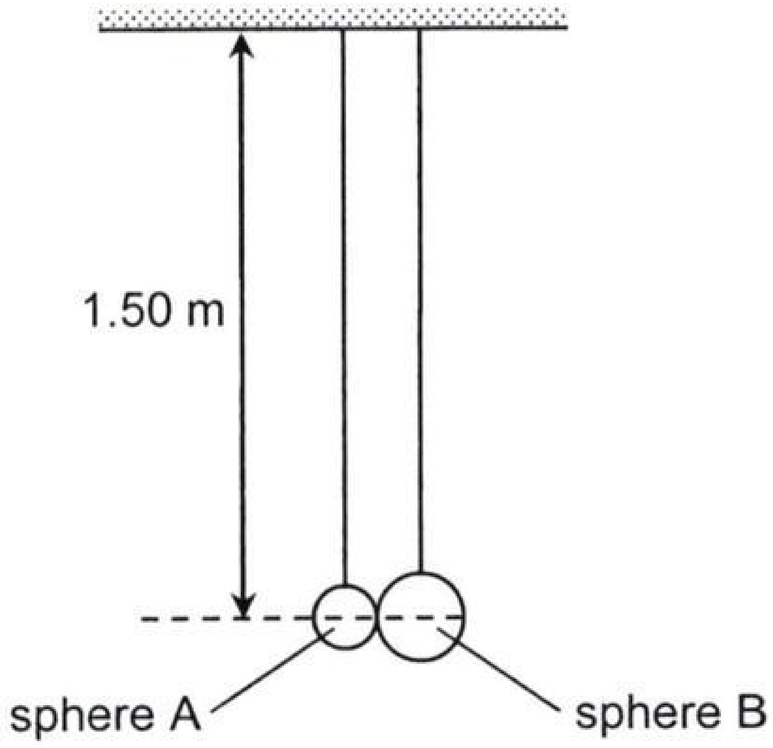


Fig. 1.1 (not to scale)

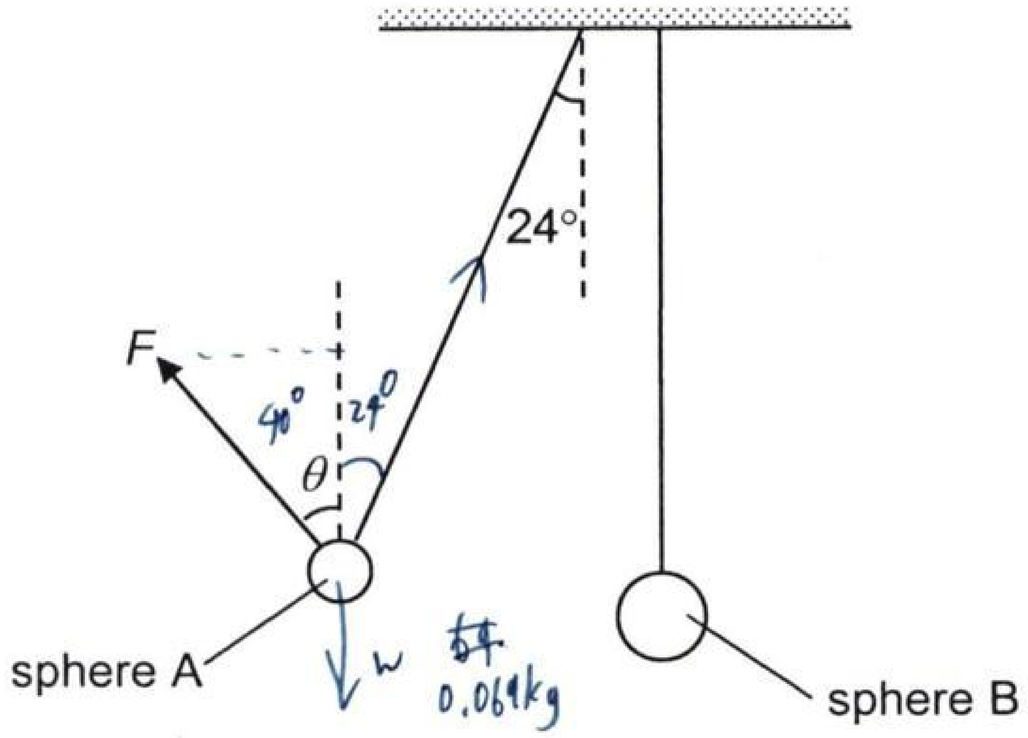
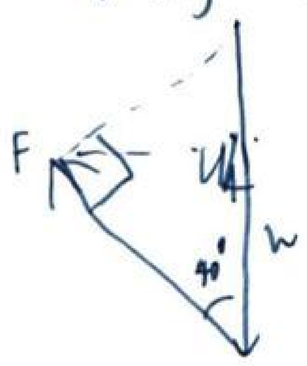


Fig. 1.2 (not to scale)

(a) (i) Calculate the magnitude of the force F required to hold sphere A in equilibrium when θ is 40°.

0 / 3

weight of A = (0.069)(9.81) = 0.6769



F = (0.6769) cos(40°)
= 0.5185
= 0.52 N (2 s.f.)



F = 0.52 N [3]

(ii) By considering the vector triangle for the forces acting on sphere A, deduce the value of θ for which the force F required to maintain equilibrium is a minimum.

0 / 1

at 90°, smallest F

θ = 90 ° [1]

(b) The force F is removed and sphere A is released from rest. It subsequently collides with sphere B.

(i) Determine the work done by the gravitational force on sphere A, from the point of release to the instant just before collision.

0 / 1

work done by the gravitational force = ... J [1] × (0)

(ii) Hence, show that the momentum of sphere A just before collision is 0.11 kg m s^{-1}.

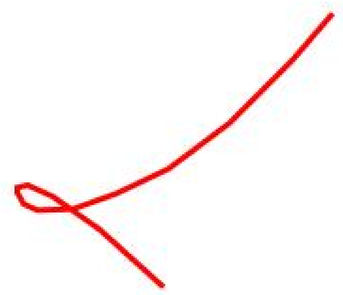
0 / 1 W =

[1] × (0)

(c) After the collision, sphere A rebounds in the opposite direction while retaining 52% of its kinetic energy just before impact. The remainder of its kinetic energy is either transferred to sphere B or transformed into other forms of energy.

(i) Calculate the momentum of sphere B after the collision.

0 / 3



momentum = ... kg m s^{-1} [3]

(ii) A student states that if the collision were such that sphere A retains none of its initial kinetic energy, more kinetic energy would be transferred to sphere B.

0 / 2

Comment on the validity of this statement.

.....

.....

.....

.....

..... [2]

× (0)

[Total: 11]

2 The radius of the Earth is 6.4×10^{6} m and the mass of the Earth is 6.0×10^{24} kg.

A spacecraft of mass 1200 kg is in a circular orbit at an altitude of 400 km above Earth's surface.

(a) Calculate the orbital speed of the spacecraft.

2 / 2

gravitational accel = centripetal accel

$r = 6.4\times10^{6} + 400\times10^{3} = 6.8\times10^{6}$ m

$$\frac{GM}{r^2} = \frac{v^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$= \sqrt{\frac{(6.67\times10^{-11})(6.0\times10^{24})}{6.8\times10^{6}}}$$

$= 7672$ m

$= 7700$ m s^{-1} (2 s.f.)

orbital speed = 7700 m s^{-1} [2]

(b) Calculate the gravitational potential energy of the spacecraft.

2 / 2

$$G.P.E = -\frac{GMm}{r}$$

$$= -\frac{(6.67\times10^{-11})(6.0\times10^{24})(1200)}{6.8\times10^{6}}$$

$= -7.062\times10^{10}$

$= -7.1\times10^{10}$ J (2 s.f.)

gravitational potential energy = −7.1 ×10^{10} J [2]

(c) Determine the additional energy required for the spacecraft to just reach infinity.

Explain your working.

2 / 3

Spacecraft in its orbit currently has $v^2 = \frac{GM}{r}$ from (a).

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{r}$$

Energy required to reach infinity is $-\frac{GMm}{r}$ from (b).

M to reach infinity, spacecraft still needs $\left|-\frac{GMm}{r} + \frac{1}{2}\frac{GMm}{r}\right| = \frac{GMm}{2r}$ J of energy

additional energy $= \frac{1}{2}\left(-\frac{GMm}{r}\right)(-1)$

$= \frac{7.062\times10^{10}}{2}$ from (b).

$= 3.531\times10^{10}$

$= 3.5\times10^{10}$ J (2 s.f.)

energy = 3.5 ×10^{10} J [3]

(d) The total energy for the spacecraft in orbit at any orbital radius r is given by

$$-\frac{GMm}{2r}$$

where M is the mass of Earth and m is the mass of spacecraft.

The spacecraft is now moved to a circular orbit closer to Earth.

(i) State the changes in the kinetic energy and gravitational potential energy of the spacecraft.

0 / 1

kinetic energy will increase but gravitational potential energy will increase at twice the increase of kinetic energy

[1]

(ii) State and explain whether positive or negative work must be done on the spacecraft.

2 / 2

Negative work done. The total amount of energy decreases and negative work is done to do so.

[2]

[Total: 10]

3 A trolley of mass m is held on a smooth horizontal surface by means of two identical springs, each of force constant k and unstretched length L. The springs are attached to fixed points as shown in Fig. 3.1.

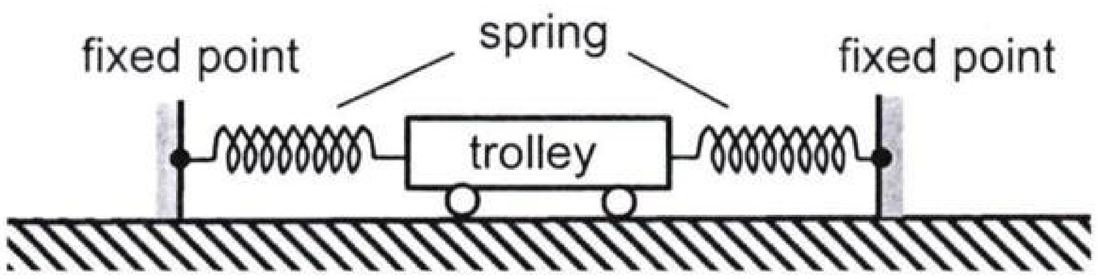


Fig. 3.1

(a) When the trolley is in equilibrium, the extension of each spring is e.

(i) The trolley is displaced horizontally to the right and then released at time $t = 0$. The trolley then oscillates with simple harmonic motion. The springs remain stretched throughout the oscillation.

0 / 2

Show that the acceleration a of the trolley is given by

$$a = -\frac{2k}{m}x$$

where x is the displacement of the trolley from its equilibrium position.

Explain your working.

by newton's second law, $F = ma$

hooke's law, $F = -kx$

~~both springs experience compression and springback by turn~~

both springs stretch and bounce

$ma = -kx$ (2) ✓

$a = -\frac{2k}{m}x$ (shown)

[2]

(ii) The mass of the trolley is 0.63 kg and the force constant of each spring is 50 N m^{-1}.

1. Determine the period T of the oscillations.

2 / 2

$k = 50$ $m = 0.63$

$a = -\omega^2 x$

$\omega = \frac{2\pi}{T}$

from (i), $\omega^2 = \frac{2k}{m}$

$\frac{2\pi}{T} = \sqrt{\frac{2k}{m}}$

$T = \frac{2\pi}{\sqrt{\frac{2(50)}{0.63}}}$

$= 0.4987$

$= 0.50$ s (2 s.f.)

$T =$ ..... 0.50 ..... s [2]

2. Determine the maximum speed of the trolley if the trolley is displaced 4.0 cm from its equilibrium position before released.

1 / 1

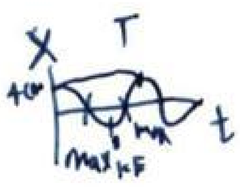
$v_{max} = \omega x_0$

$= \sqrt{\frac{2(50)}{0.63}}\,(4.0 \times 10^{-2})$

$= 0.50$

$= 0.50$ (2 s.f.) m s^{-1}

maximum speed = ... 0.50 ..... m s^{-1} [1]



3. On Fig 3.2, sketch a graph to show the variation with t of the kinetic energy E_k of the trolley for one complete oscillation, starting at $t = 0$.

Include all relevant values.

max k_E @ max v

1 / 2

$\frac{1}{2}(0.63)(0.50) = 0.1575$

$= 0.16$ (2 s.f.)

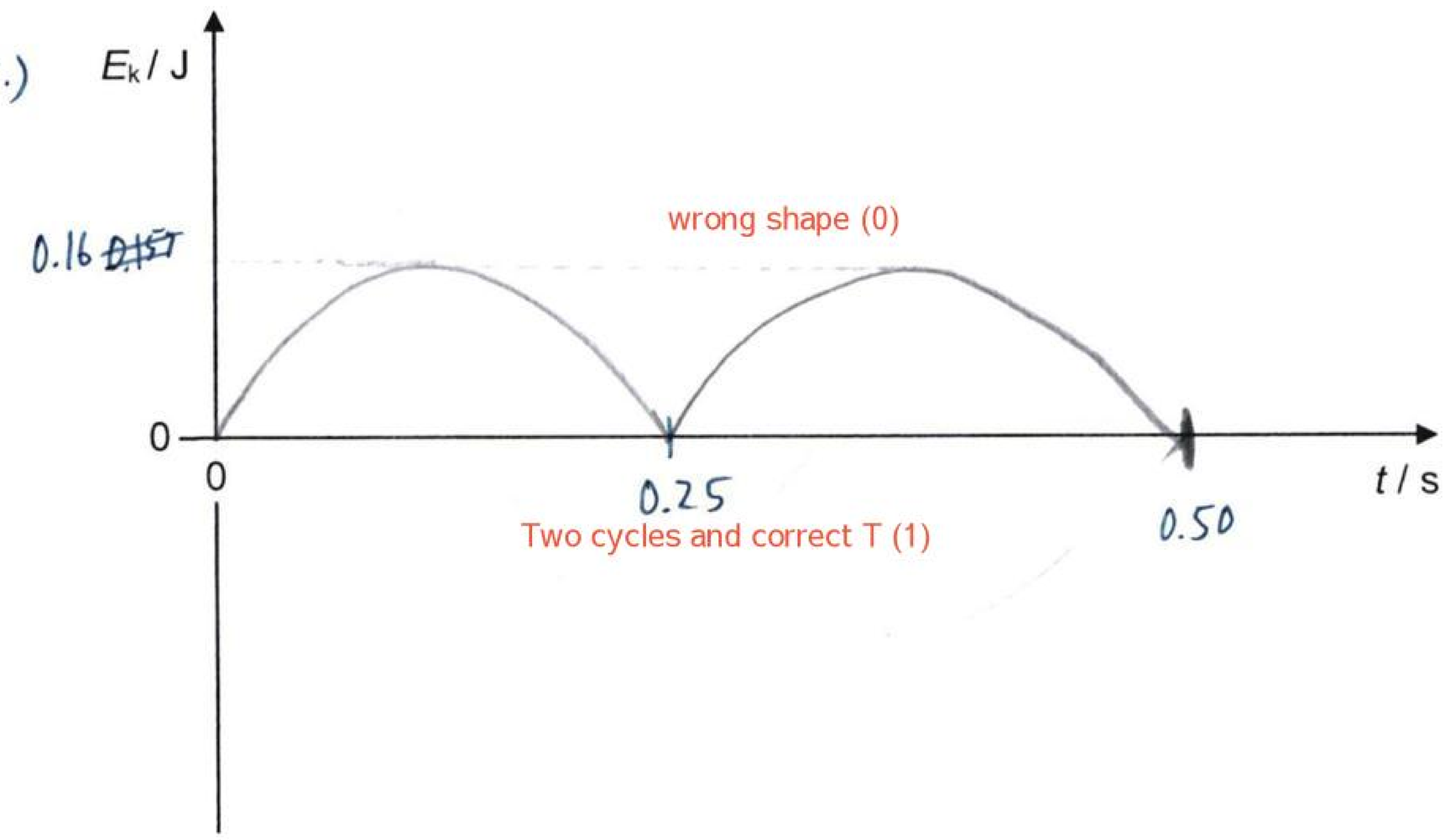


Fig 3.2

[2]

(b) One additional identical spring is added on each side of the trolley as shown in Fig 3.3. The trolley is then made to oscillate.

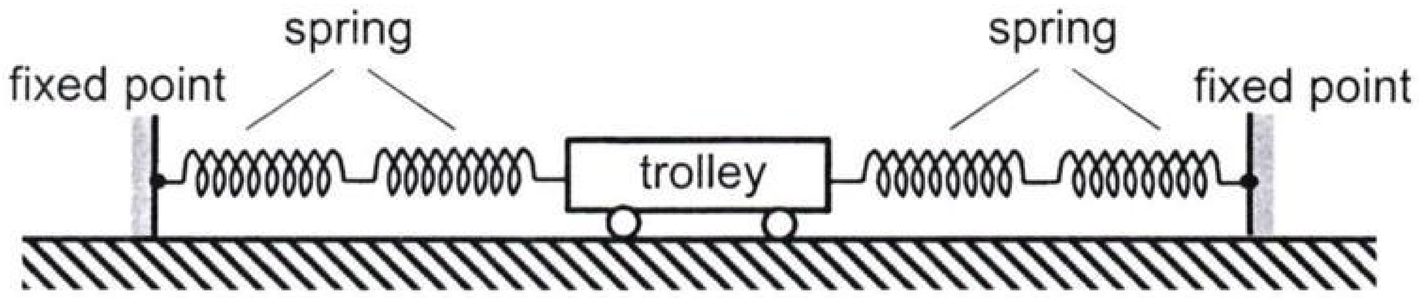


Fig 3.3

State and explain the change, if any, in the period of oscillation of the trolley.

0 / 2

Since the springs are identical, there is no change in the behaviour of oscillation. The spring constant k remains the same and the displacement is still the same, so none of the factors affecting oscillation have been changed.

[2]

[Total: 9]

4 **(a)** An incoming polarised beam of monochromatic light from a light source passes through two plane polarisers. I_0 and I_1 are the intensities of the light along the central axis just before and after passing through the second polariser respectively as shown in Fig. 4.1.

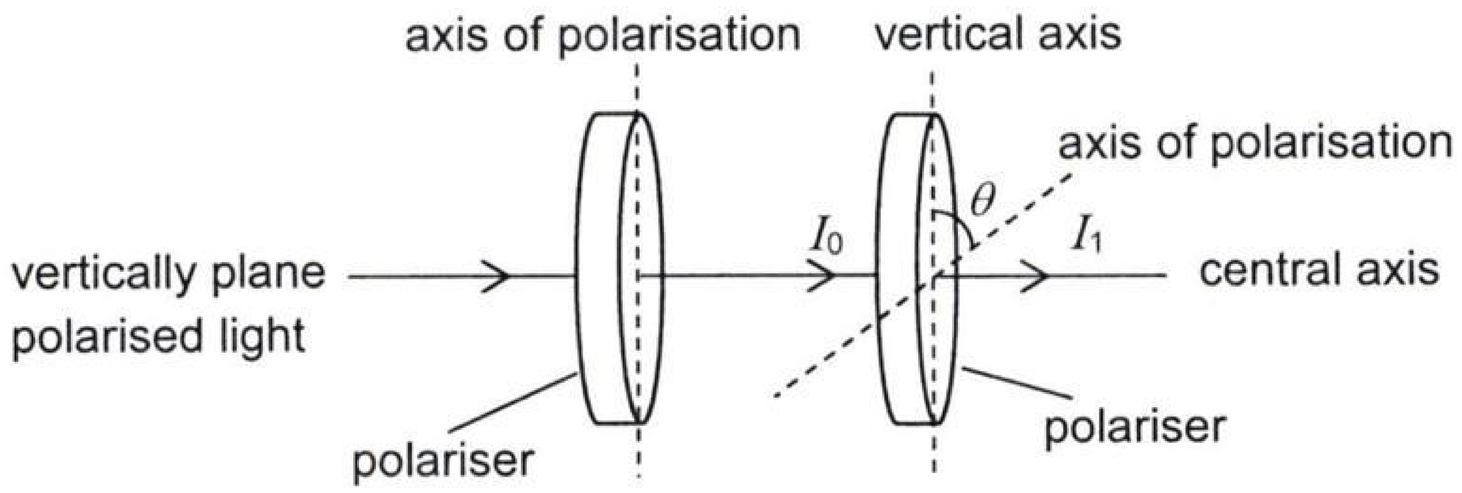


Fig. 4.1

The incoming light is polarised in the vertical plane. The two polarisers have their axis of polarisation at an angle θ to one another and the variation of I_1 with $\cos^2\theta$ is shown in Fig. 4.2.

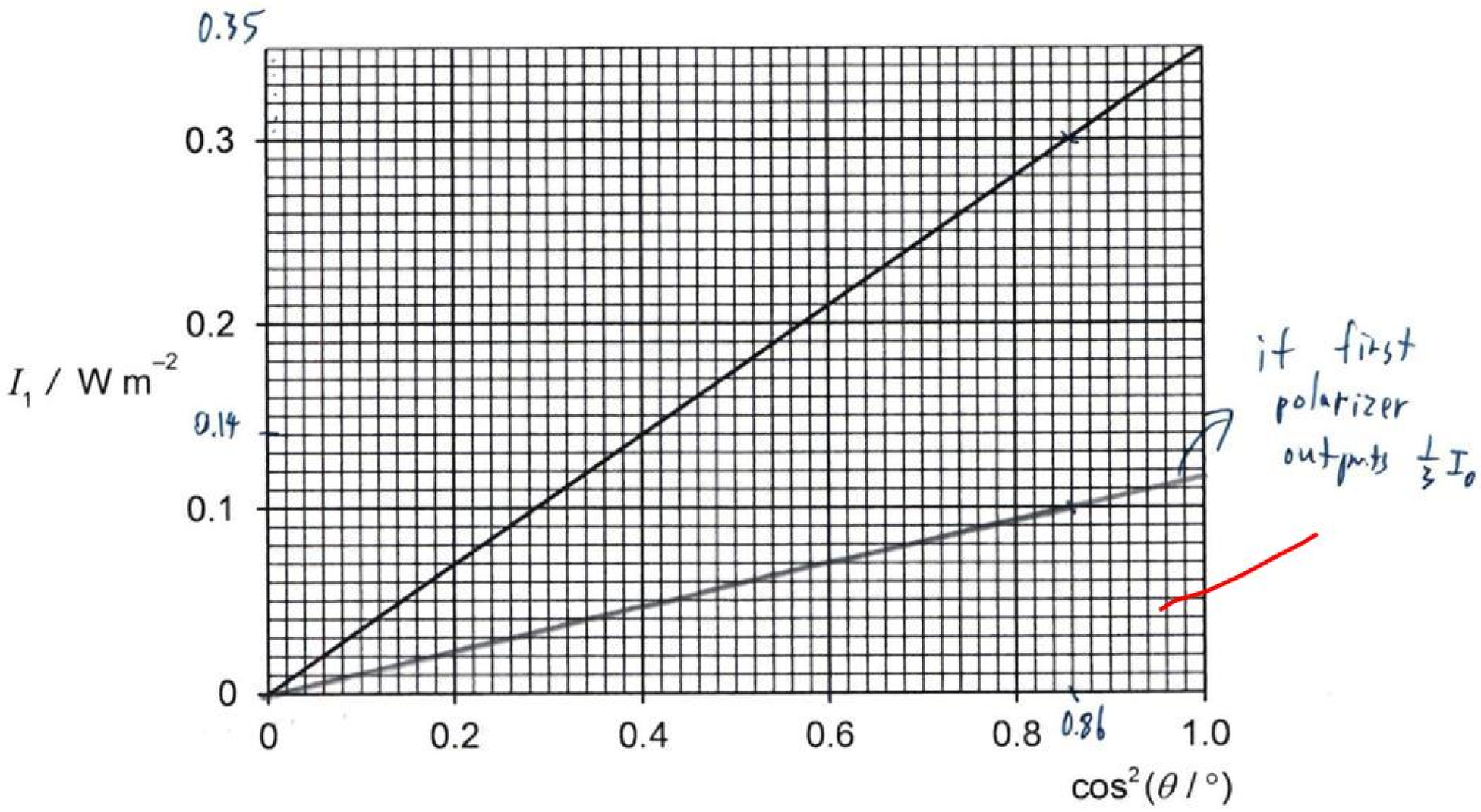


Fig. 4.2

(i) Determine two possible values of θ such that $\frac{I_1}{I_0} = 0.50$ where $0° < \theta < 360°$.

2 / 2

for $\cos^2(\theta) = \frac{1}{2}$,

$\theta = 45°$ or $135°$

clearer working

$\theta =$ 45 ° and 135 ° [2] 2

(ii) The light source is adjusted such that the intensity of the light just before passing through the second polariser is now $\frac{1}{3}I_0$.

Draw a line in Fig. 4.2 to indicate the new variation of the intensity I_1 with $\cos^2\theta$.

1 / 1 [1]

(b) The polarisers are now removed and a light intensity detector is placed at different horizontal distances from the light source which is a point source of light.

original is $2I_0$

The detector indicates a value of 0.20 W m^{-2} when it is at a distance of 20 cm from the light source.

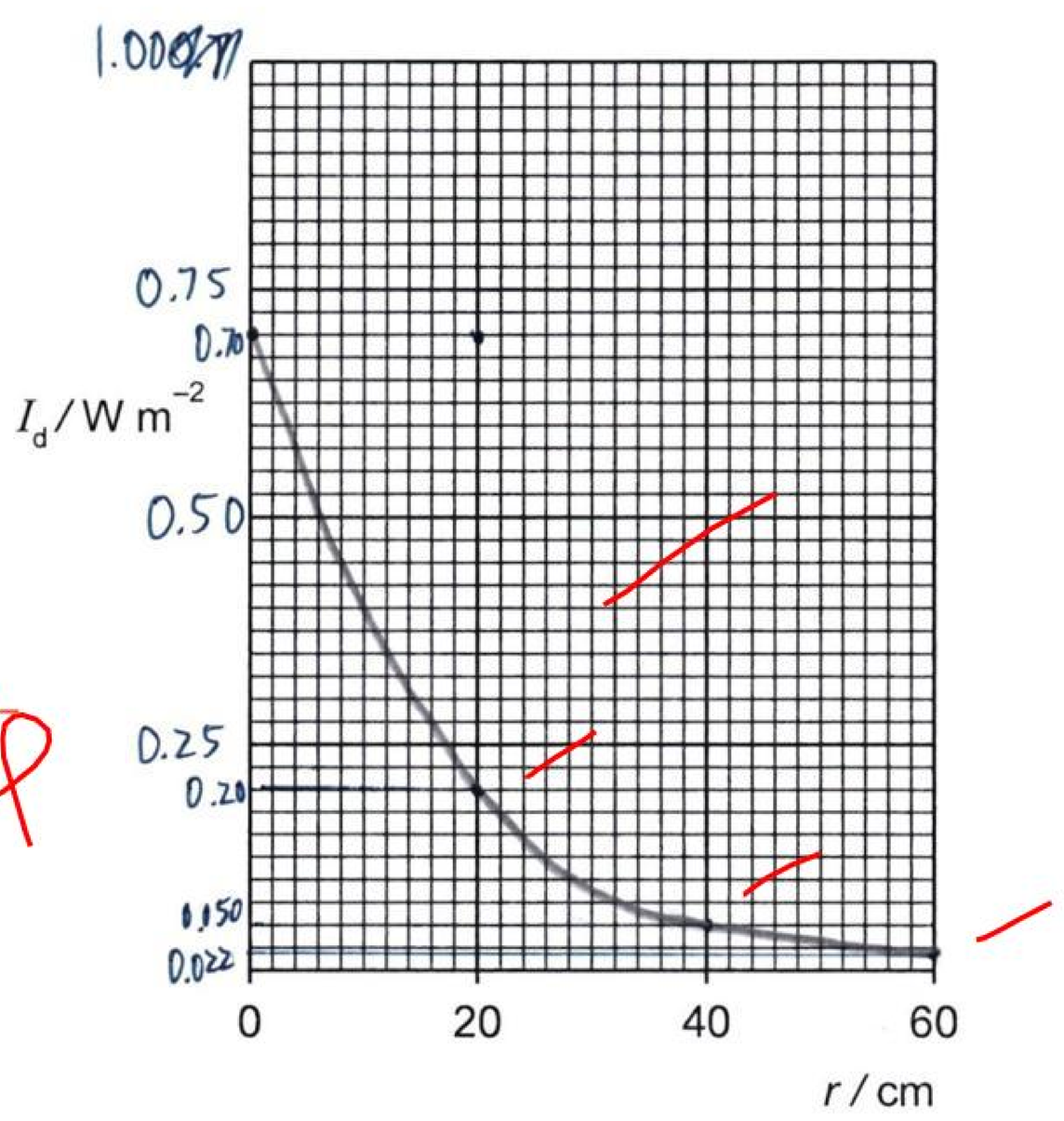
On the axes of Fig. 4.3, sketch a graph to show the variation of the intensity I_d measured by the detector with distance r from the source. The graph should extend from $r = 20$ cm to $r = 60$ cm.

Include an appropriate scale for the intensity axis.

$I_0 = 0.35$

e.g. is $(0.35)(2) = 0.7$

2 / 3 $I_1 \propto \frac{1}{r^2}$



@ 40 cm, $I_1 = 0.050$

@ 60 cm, $I_1 = 0.022$

Inappropriate scale (−1)

Fig 4.3

[3]

[Total: 6]

how many times has this been tested?

5 A capacitor of capacitance C is connected in parallel with a resistor of resistance 200 Ω across an alternating supply as shown in Fig. 5.1.

$C = \frac{Q}{V}$

$U = \frac{1}{2}QV$

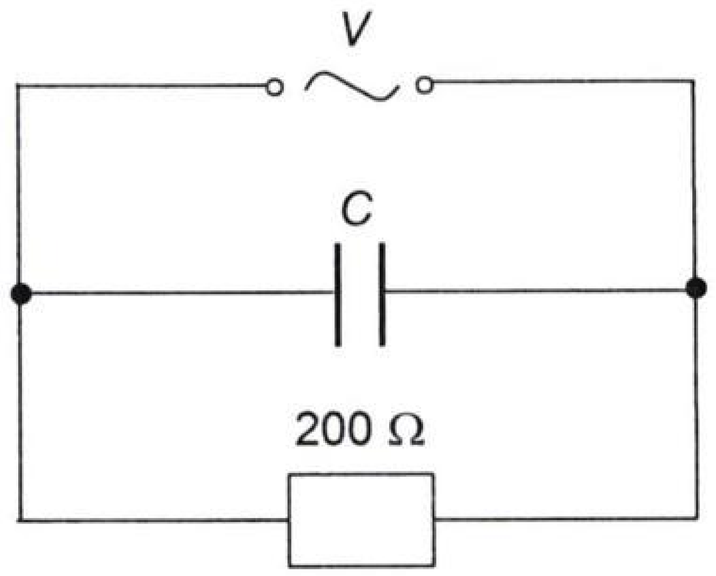


Fig. 5.1

The variation with time t of the potential difference V of the alternating supply is given by

$$V = 120\sin(314t),$$

where V is measured in volts and t in seconds.

(a) (i) Write an equation, in terms of C, that relates how charge Q on the capacitor plate varies with t.

1 / 1

$C = \frac{Q}{V}$

$Q = CV$

$= (120\sin(314t))$ ✓

[1]

(ii) Hence, show the current I_C through the capacitor is $37700C\cos(314t)$.

1 / 1

$I_C = \frac{dQ}{dt}$

$= (120)(314)(\cos(314t))$

$= 37680(\cos(314t))$ ✓

$= 37700(\cos(314t))$ (3 s.f.)

[1]

(b) (i) Determine the r.m.s. current $I_{R,rms}$ through the resistor.

1 / 1

$I_{krms} = \frac{V_{rms}}{R} = \frac{\frac{\sqrt{2}}{2}(120)}{200}$

$= 0.42426$

$= 0.424$ A (3 s.f.)

$I_{R,rms}$ = 0.424 ✓ A [1]

(ii) Calculate the mean power P dissipated in the resistor.

0 / 2

$P = I^2R$

~~IAV~~

$\langle P \rangle = I_{rms}^2 R$

$= (0.42426^2)(200)$

$= 84.852$

$= 84.9\ W$ (3 s.f.)

$P =$ 84.9 W [2]

(c) The frequency of V is increased.

State and explain the effect, if any, on

(i) the r.m.s. current through the resistor,

1 / 1

No effect. r.m.s voltage is dependent on amplitude and shape of alternating current, not frequency, so current which is $\frac{V_{rms}}{R}$, has no change. [1]

(ii) the r.m.s. current through the capacitor.

0 / 1

[1]

[Total: 7]

6 Fig. 6.1 shows the current-voltage (I-V) characteristics of a resistor and a diode.

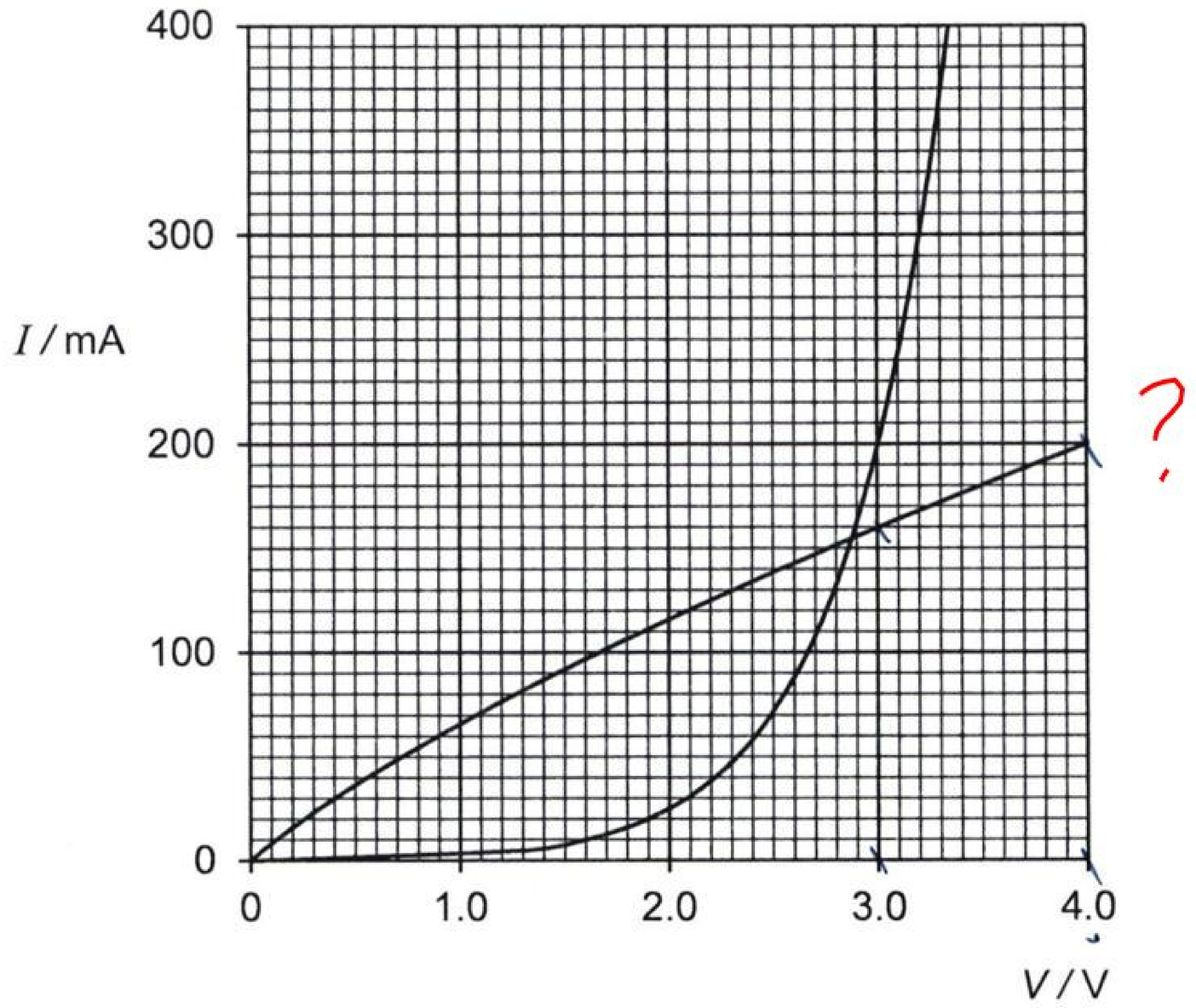


Fig. 6.1

When the resistor and diode are connected in series with a battery of negligible internal resistance, the current in the circuit is 200 mA.

(a) Determine

(i) the e.m.f. of the battery,

@ 200 mA

2 / 2

P.D. across both components = e.m.f. = 3.0 + 4.0
= 7.0 V
– 2 dp.

e.m.f. = 7.0 V [2]

sf/dp

0 / 1

(ii) the resistance of the diode.

$V = IR$

$R = \frac{V}{I}$

$= \frac{4.0}{40200 \times 10^{-3}}$ ✗

$= 20\ \Omega$

resistance = ... 20 ✗ ..... Ω [1]

(b) (i) On Fig. 6.1, sketch the *I-V* characteristic of a negative temperature coefficient thermistor. Exact values are not required. [1]

0 / 1

Not drawn.

(ii) Explain, in microscopic terms, the difference in the *I-V* characteristics between the thermistor and the resistor.

0 / 1

.....

.....

..... [1]

[Total: 5]

7 A proton travelling in a vacuum at a speed of 2.0×10^5 m s^{-1} enters a region of uniform magnetic field of magnetic flux density 3.5 mT . The magnetic field is directed into the plane of the paper.

Initially, the proton is travelling in a direction normal to the magnetic field. The path of the proton while in the magnetic field is the arc of a quarter circle as shown in Fig. 7.1.

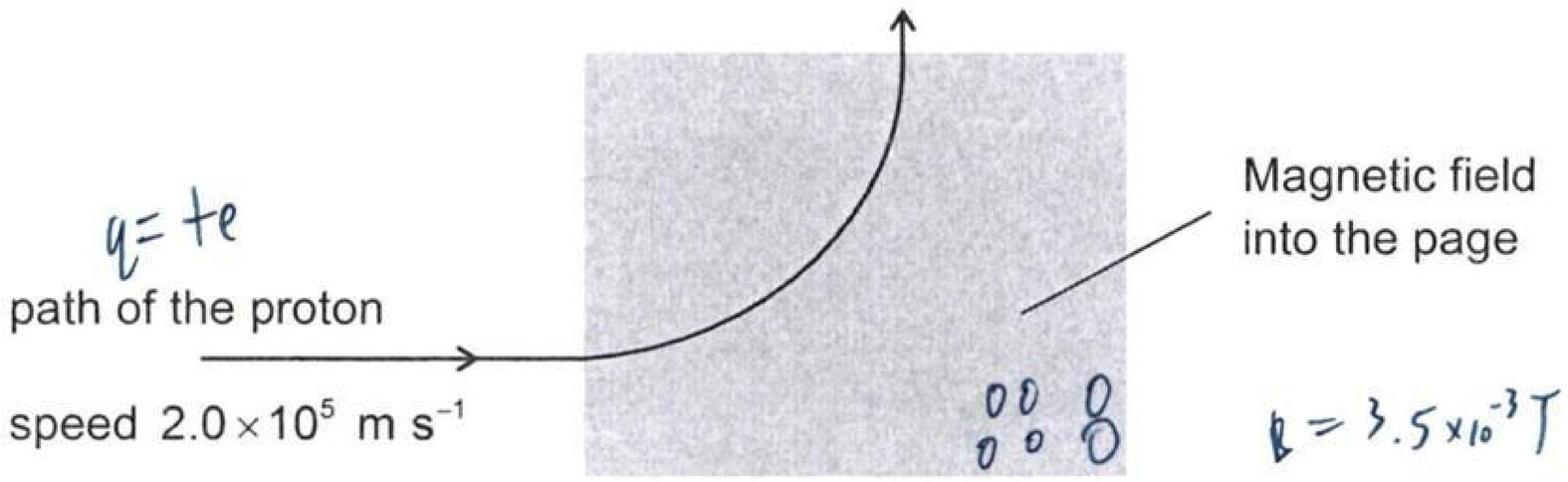


Fig. 7.1

(a) Explain why the path in the magnetic field is an arc of a circle.

0 / 2

Using fleming's left hand rule, we can deduce that the path of the proton that is normal to the magnetic field produces a magnetic force acting on it perpendicular to both the magnetic field and path of proton. The force that is perpendicular to the motion of the proton results in the circular motion of the proton, causing it to arc in a quarter circle until exit fon. [2]

missing key points (-2)

(b) (i) Show that the radius of the circular arc is 0.60 m .

1 / 1

$F = Bqv$ provides centripetal force $\frac{mv^2}{r}$

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

$$= \frac{(1.67\times10^{-27})(2.0\times10^5)}{(3.5\times10^{-3})(1.60\times10^{-19})}$$

$$= 0.5964$$

$$= 0.60\,\text{m (2 s.f.) (shown)}$$

[1]

(ii) By considering the change in momentum of the proton, calculate the magnitude of the average force exerted on the proton while it was in the uniform magnetic field.

4 / 4

by newton's second law, $F = \frac{\Delta p}{\Delta t}$

mass is the same, so $\Delta p = m \Delta V$

in circular motion,



$\Delta V = 2V \sin\frac{\Delta\theta}{2}$

$\Delta p = 2mV \sin\frac{\Delta\theta}{2}$

rate of change in angle $\frac{\Delta\theta}{\Delta t} = \omega = \frac{V}{r}$

in quarter circle, $\Delta\theta = 90^0$

$\Delta p = 2mV \sin\frac{90^0}{2}$

$= \sqrt{2}\, mV$

$\Delta t = \frac{\text{distance}}{V} = \frac{\frac{1}{2}\pi r}{V} = \frac{\frac{1}{2}(\pi)(0.[illegible]5464)}{2.0 \times 10^5}$

$= 4.684 \times 10^{-6}$ s

$F = \frac{\sqrt{2}\, mV}{4.684 \times 10^{-6}} = \frac{\sqrt{2}(1.67 \times 10^{-27})(2.0 \times 10^5)}{4.684 \times 10^{-6}}$

$= 1.008 \times 10^{-16}$

$= 1.0 \times 10^{-16}$ N

average force = 1.0 x 10^-16 N [4]

[Total: 7]

End of Section A

Section B

Answer **one** question in this section in the spaces provided.

8 **(a)** Define *magnetic flux density*.

0 / 1

.....

.....

..... [1]

(b) **(i)** Fig. 8.1 shows a flat circular coil carrying a current in the direction shown.

1 / 3

Sketch the magnetic field pattern due to the current in the coil within the dashed region.

more than 2 parallel lines in the middle

lines are not equally spaced

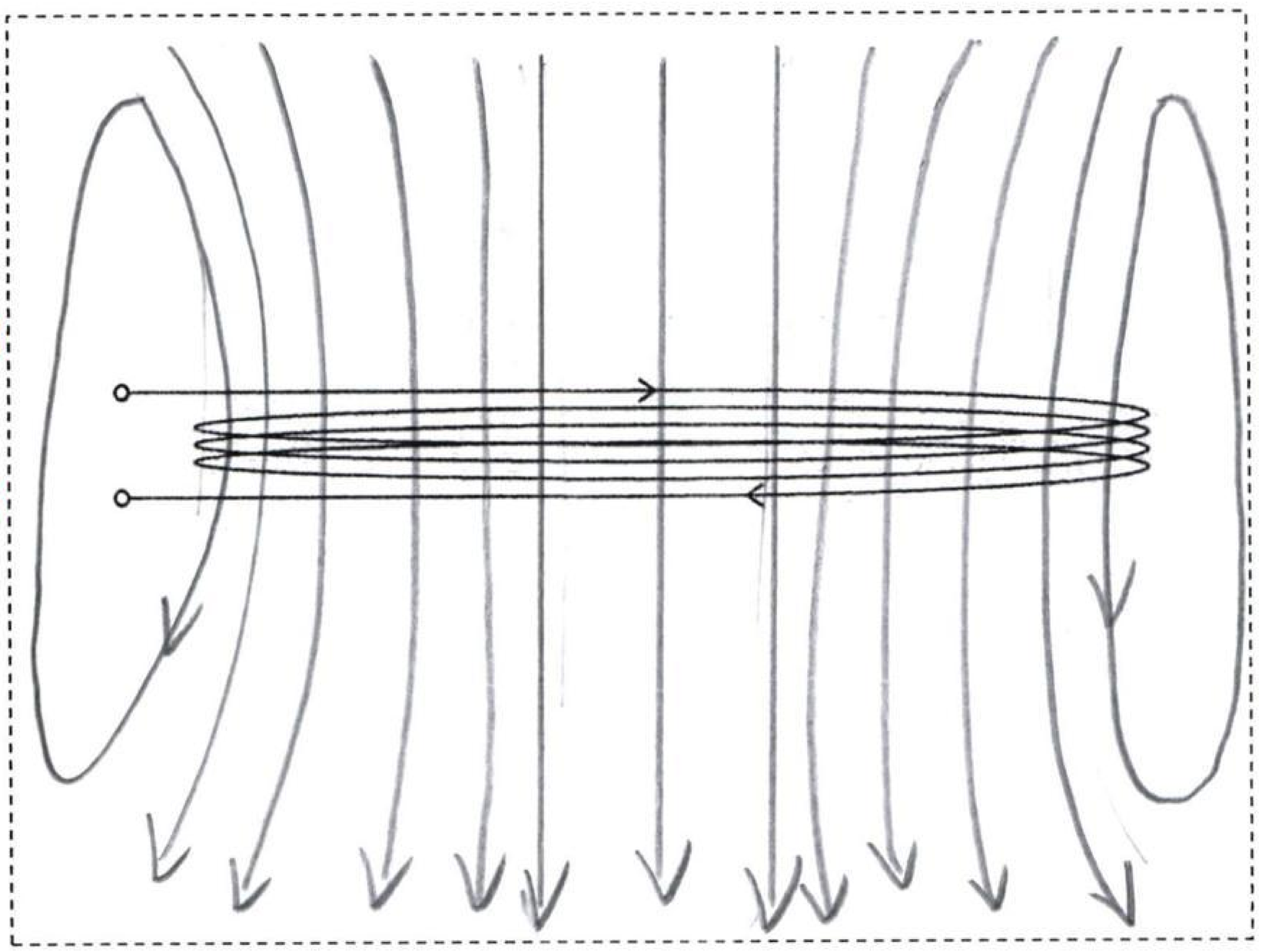


Fig. 8.1 (not to scale)

[3]

(ii) The flat circular coil in **(b)(i)** is fixed in position. A straight wire carrying current is then placed near the flat circular coil.

Explain the effect on the straight wire when it is placed

1. above the coil as shown in Fig. 8.2,

2 / 2

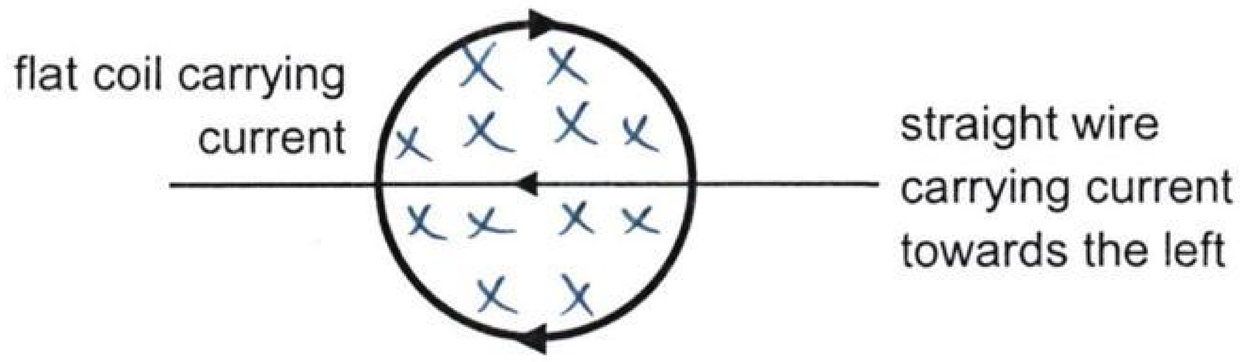


Fig. 8.2 (top view)

Wire will move downwards. Using flaming's left hand rule, force on wire points downwards from the magnetic field pointing into the page derived with right hand grip rule, and the current of wire flowing towards the left.

2. along the axis of the coil, as shown in Fig 8.3.

1 / 1

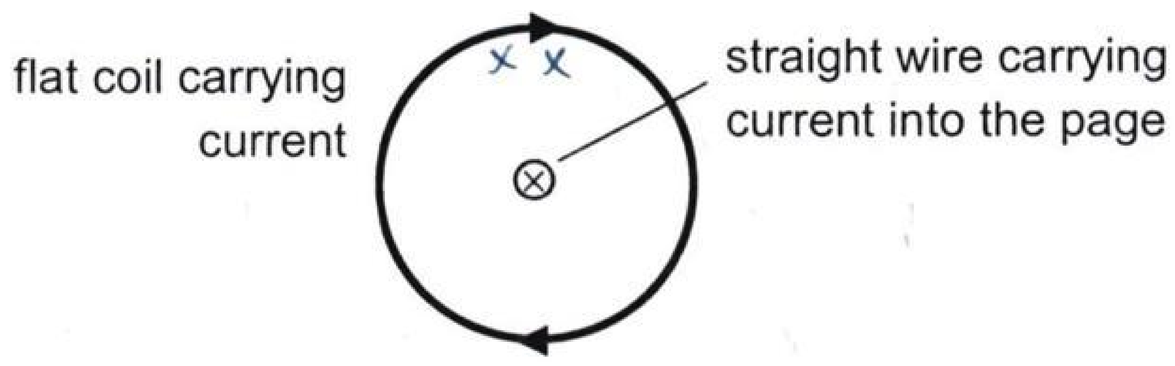


Fig. 8.3 (top view)

No effect. Current is parallel to magnetic field, which results in no component of both being perpendicular to each other and there resulting in 0 force on the wire.

[3]

(c) Fig. 8.4 shows a metal saucepan placed on an electromagnetic induction cooking hob.

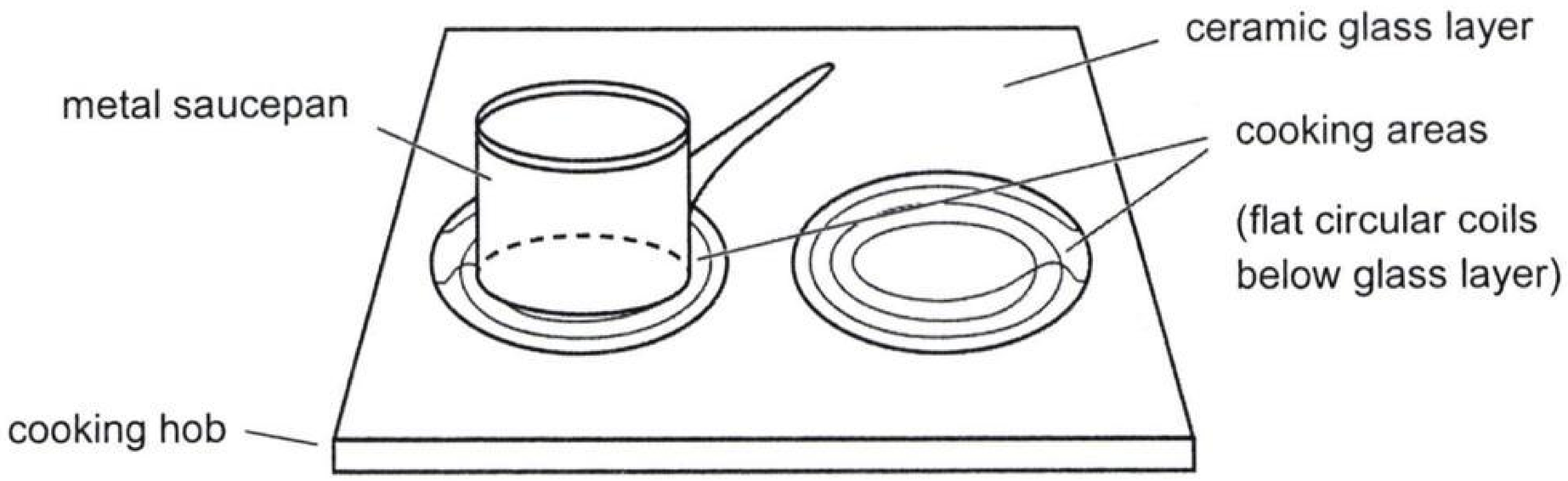


Fig. 8.4

Each cooking area of the induction cooking hob consists of a flat circular coil with 20 turns below a layer of ceramic glass. When switched on, the coils carry a high frequency alternating current.

(i) Explain how a high frequency alternating current in the coils results in a significant transfer of energy to the contents in the saucepan.

3 / 3

By Faraday's law of electromagnetic induction, the induced e.m.f. is ~~the~~ proportionate to rate of change in magnetic flux linkage. When the coils alternate in current at a high frequency, the magnetic ~~flux linkage of the~~ ~~coils~~ fields of the coils constantly alternates at the same frequency. This produces a rapidly changing magnetic field, ~~it~~ which in turn rapidly changes the magnetic flux linkage of the saucepan. The ~~w~~high rate of change in magnetic flux linkage induces a large, significant amount of eddy currents in the saucepan, which ends up transferring the heat ~~lost in the~~ from the eddy currents to the contents in the saucepan. [3]

emf?

(ii) Fig. 8.5 shows the variation with time t of the alternating current I_{coil} flowing through the coil.

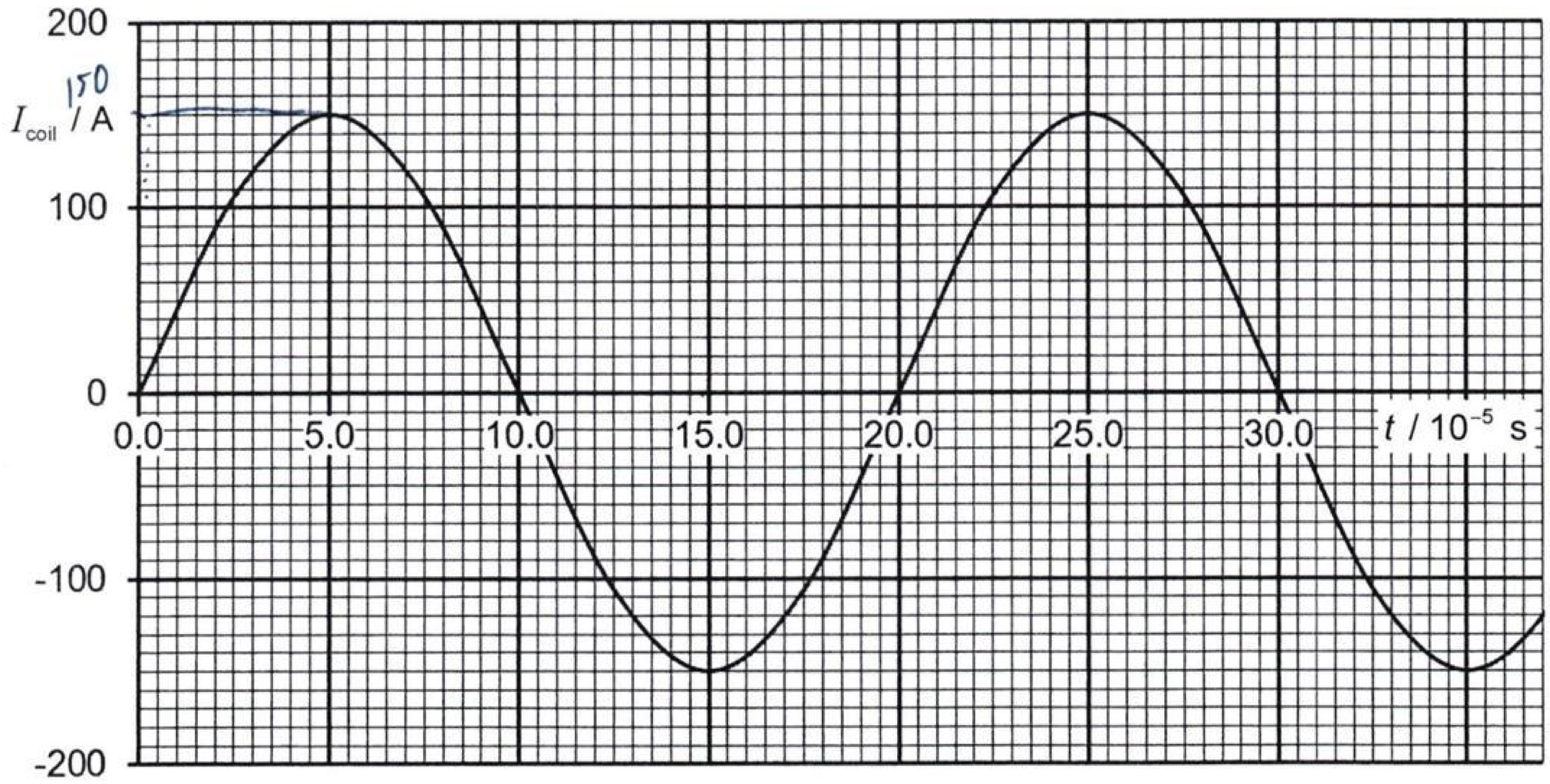


Fig. 8.5

The metal saucepan and its contents have a total mass of 900 g and an effective specific heat capacity of 2000 J kg^{-1} K^{-1}. The circular base of the saucepan has an effective area and resistance of 0.020 m^2 and 0.20 Ω respectively.

The magnetic flux density throughout the base of the saucepan can be taken to be $5\mu_o NI_{coil}$ where μ_o is the permeability of free space and N is the number of turns in the flat circular coil.

Assume there is no loss of energy between the coil and the saucepan.

1. Use Fig 8.5 to determine the maximum value of the induced current in the base of the saucepan.

1 / 3

amplitude = 150
period = 20 s

$I_c = 150 \sin\left(\frac{2\pi}{20}t\right)$

$\frac{dI_c}{dt} = 3000\pi \cos\left(\frac{2\pi}{20}t\right)$

$\left.\frac{dI_c}{dt}\right|_{t=10} = 3000 \cos\left(\frac{2\pi}{20} \times 10\right)$

$= 2995$

$= 3000$

induced emf$_{max} = \frac{\Delta\Phi}{\Delta t} = \frac{dB}{dt} A$

$= \frac{dI}{dt} (5\mu_0 N A)$

$= (3000)(4\pi \times 10^{-7})(5)(4\pi \times 10^{-7})(0.020)(20)$

$= 7.5340 \times 10^{-3}$

max induced current $= \frac{V}{R} = \frac{7.540 \times 10^{-3}}{0.2}$

$= 0.03770$

$= 0.038$

maximum value of induced current = ... 0.038 ..... A [3]

0 / 3

2. Sketch, on Fig. 8.6, a graph to show the variation with t of the ~~rate of transfer of energy from the coil to~~ the saucepan for the first 30×10^{-5} s. Label all relevant values.

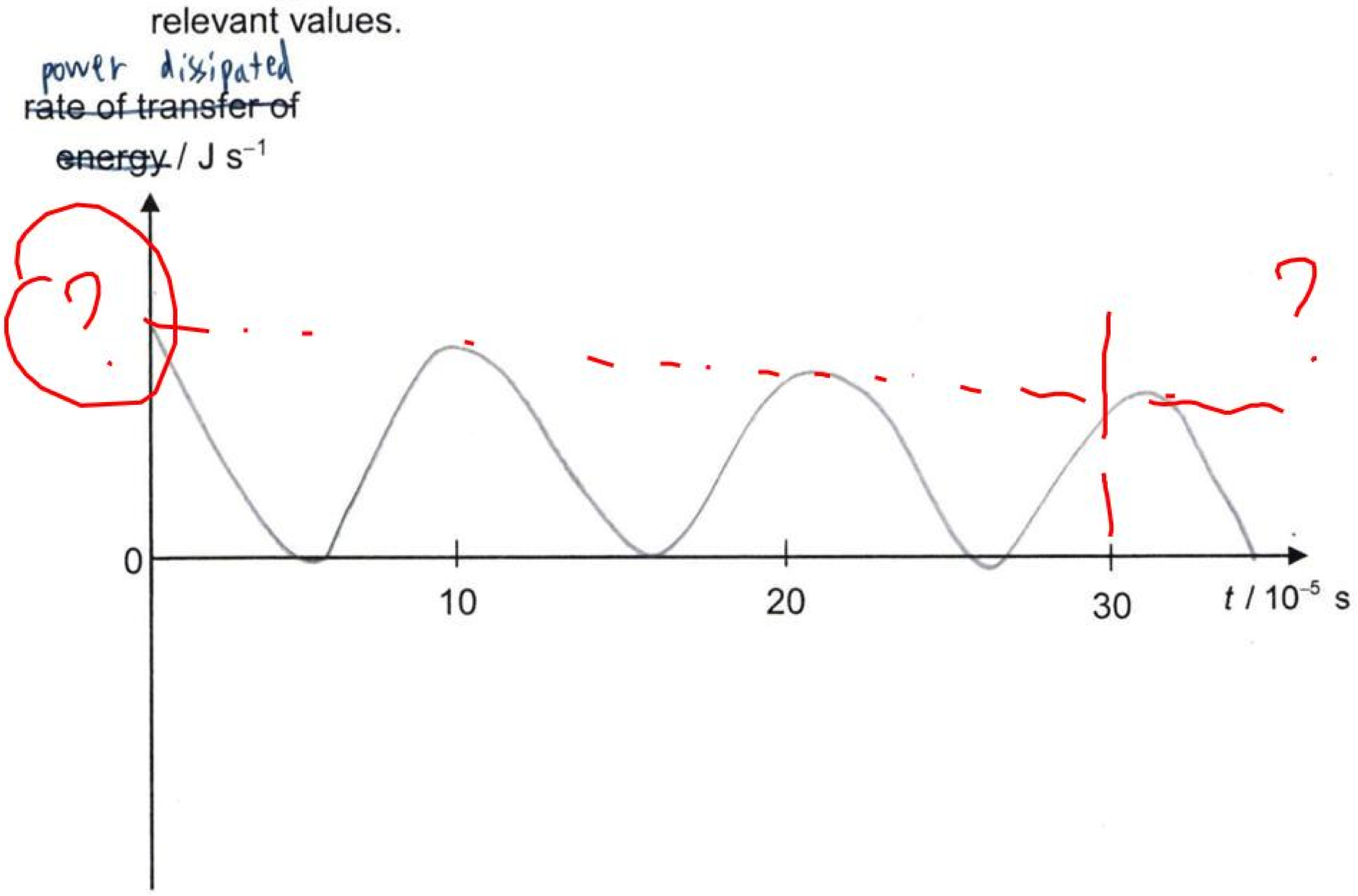


Fig. 8.6

[3]

3. Calculate the temperature rise of the saucepan and contents after 4.0 minutes of use.

0 / 3

temperature rise = ... ° C [3]

(iii) Stainless steel has a high magnetic permeability. Explain why stainless steel cookware are preferred in induction cooking.

0 / 1

This allows for more efficient and faster heat transfer. Why?

.....

..... [1]

[Total: 20]

9 **(a)** State how a stationary wave is formed.

.....

.....

.....

..... [2]

(b) A stationary wave is set up in a stretched string. The variation of the maximum vertical displacement *y* of the string with distance *x* along the stationary wave is shown in Fig. 9.1.

P, Q and R are points on the string. The time taken by point P to travel from its maximum upward displacement to its maximum downward displacement is 9.0×10^{-3} s.

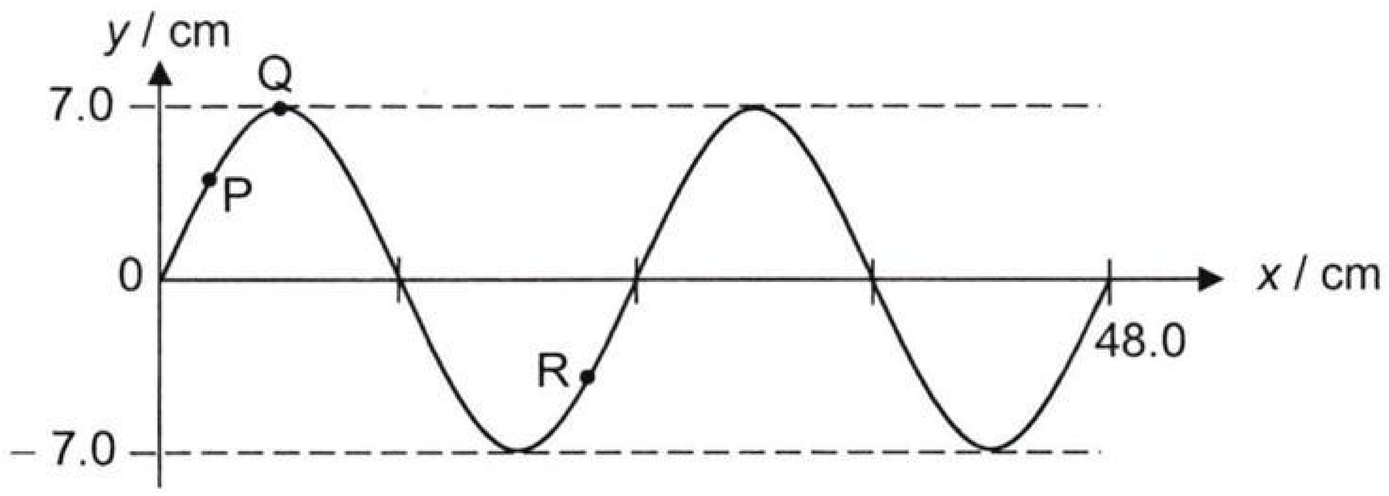


Fig. 9.1

(i) Determine, for each wave which formed the stationary wave, its

1. amplitude,

amplitude = ... cm [1]

2. frequency,

frequency = ... Hz [1]

3. speed.

speed = ... m s^{-1} [2]

(ii) State the phase difference between points

1. P and Q,

phase difference = ... rad [1]

2. Q and R.

phase difference = ... rad [1]

(c) A particle of mass *m* is trapped in a one-dimensional box of width *L* with perfectly reflecting walls. Its wavefunction is given by

$$\Psi_n(x) = \begin{cases} A\sin\left(\dfrac{2\pi x}{\lambda_n}\right) & 0 \le x \le L \\ 0 & \text{otherwise} \end{cases}$$

where λ_n is the de Broglie wavelength of the particle, *n* is the quantum number and *A* is the normalisation constant.

(i) Explain why the value of $\int_0^L |\Psi|^2 \, dx$ must be 1.

.....

.....

..... [1]

(ii) Write down an expression for λ_n in terms of L and n.

[1]

(iii) For $n = 3$, the wavefunction is $\Psi_3(x)$.

On Fig. 9.2, sketch the probability density function for the particle. Label the axes with appropriate values.

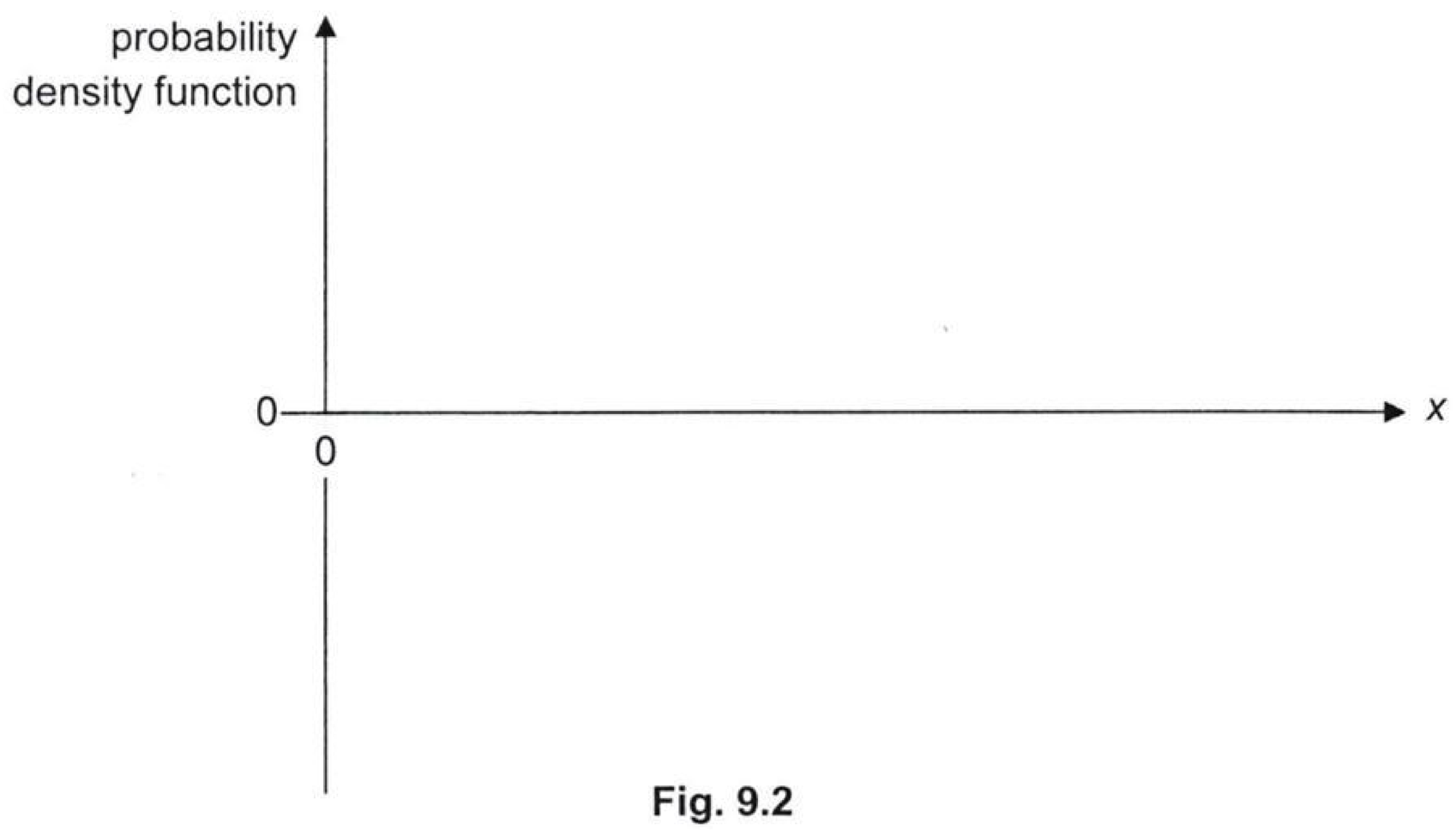


Fig. 9.2

[2]

(iv) Calculate the value of A. Show your working clearly.

[2]

(v) Suggest what might happen to the particle if the walls of the box were not perfectly reflecting.

.....

.....

..... [1]

(d) In 1980, Japanese Physicist Akira Tonomura carried out a double-slit experiment using electrons.

Fig. 9.3 shows the number of electrons that have arrived at the screen for various elapsed times *t*.

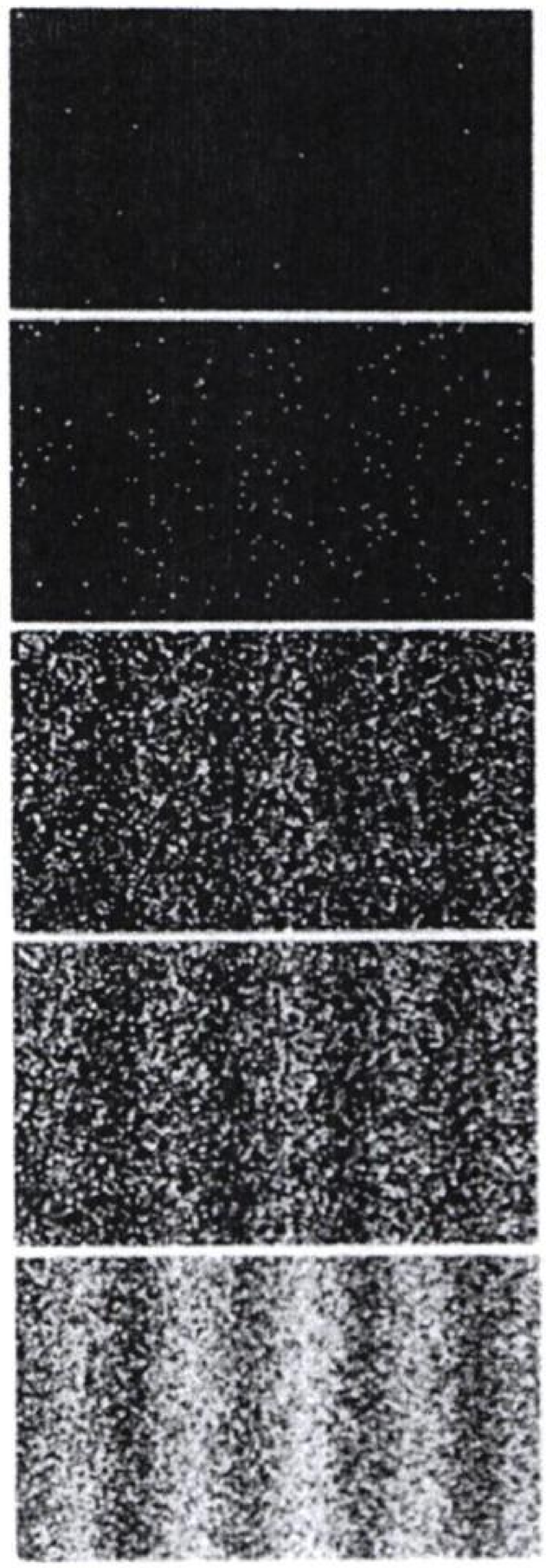


Fig. 9.3

(i) Explain how the images at various elapsed times in Fig. 9.3 show that electrons exhibit both particle-like and wave-like properties.

Particle-like: .....

.....

.....

Wave-like: .....

.....

..... [2]

(ii) The image in Fig. 9.3 shows that after an elapsed time of 20 minutes, a regular pattern has formed.

By considering the de Broglie relation, state and explain what this regularity implies about the velocities of the electrons used in the experiment.

.....

.....

.....

.....

.....

..... [3]

[Total: 20]

End of Section B